Digital Design Assignment 1

AMBA AHB – Lite Master

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AHB Master Module

Design & Verification Documentation

1. Features Supported

- FSM-based AHB master with four states: IDLE, BUSY, NONSEQ, SEQ.
- Supports HTRANS transfer types: IDLE, BUSY, NONSEQ, SEQ.
- Supports HBURST types: SINGLE and INCR (undefined-length increment).
- Supports HSIZE transfer sizes: BYTE, HALFWORD, WORD.
- Wait-state handling: master holds address/control signals and drives HTRANS = BUSY while HREADY is low.
- Fixed protection attributes: HPROT tied to 4'b0011, HMASTLOCK tied low (locked transfers not supported).

2. Overview

The master module implements the address- and data-phase generation logic for an AMBA AHB master. A 2-bit finite state machine sequences the HTRANS output between IDLE, BUSY, NONSEQ, and SEQ based on the incoming transaction request (tr) and the slave's HREADY signal. Address-phase signals (HADDR, HWRITE, HSIZE, HBURST, HTRANS) are combinational outputs of the current state, while HWDATA is registered so that it is valid during the following data phase, matching the standard AHB pipelined address/data timing.

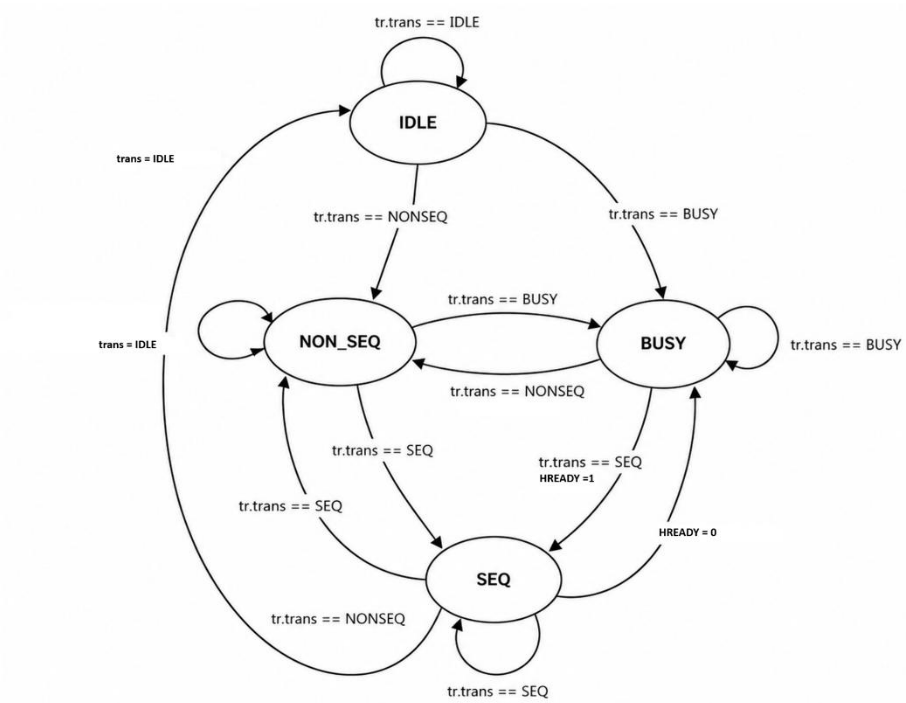
3. Module Interface

Signal	Dir	Width	Description
HCLK	in	1	AHB clock
HRESETn	in	1	Active-low asynchronous reset
HRDATA	in	32	Read data from slave (routed to testbench; not consumed internally)
HREADY	in	1	Slave ready / wait-state indicator
HRESP	in	1	Slave response (error indicator)
tr	in	struct	transaction_t request: trans, burst, size, write, addr, wdata
HADDR	out	32	Address bus
HWRITE	out	1	1 = write, 0 = read
HSIZE	out	hsize_t	Transfer size: BYTE / HALFWORD / WORD
HBURST	out	hburst_t	Burst type: SINGLE / INCR
HPROT	out	4	Protection control, fixed to 4'b0011
HTRANS	out	htrans_t	Transfer type: IDLE / BUSY / NONSEQ / SEQ
HMASTLOCK	out	1	Locked transfer indicator, tied low
HWDATA	out	32	Write data bus, registered

4. State Machine

current_state is an active-low, asynchronously reset register updated on the rising edge of HCLK. On reset it defaults to s_IDLE.

State	HTRANS driven	Next-state trigger
s_IDLE	IDLE	Moves to s_NON_SEQ when tr.trans == NONSEQ
s_BUSY	BUSY	Moves to s_SEQ / s_NON_SEQ / s_IDLE per tr.trans, else remains BUSY
s_NON_SEQ	NONSEQ	Drops to s_BUSY if !HREADY; else advances per tr.trans; returns to s_IDLE if tr.trans==NONSEQ and HBURST==SINGLE
s_SEQ	SEQ	Drops to s_BUSY if !HREADY; else advances per tr.trans



5. Testbench (test_ahb.sv)

test_ahb instantiates master as DUT and drives it through two reusable tasks:

- reset(): asserts HRESETn low for 3 clocks with HREADY=1, HRESP=0, then releases reset.
- send_transaction(trans, burst, size, write, addr, data): loads the tr struct, waits one clock edge, and prints state/HTRANS/HADDR/HWDATA/HBURST to the transcript.

Test scenarios exercised in the initial block:

#	Scenario	HTRANS / HBURST / HSIZE	Addr / Data
1	Idle cycle	IDLE / SINGLE / BYTE	0x0 / 0x0
2	Single word write	NONSEQ / SINGLE / WORD	0x1000_0000 / 0xAAAA5555
3	Incrementing byte write	NONSEQ / INCR / BYTE	0x2000_0000 / 0x11
4	Single half-word write	NONSEQ / SINGLE / HALFWORD	0x3000_0000 / 0x1234
5	Single word read + HREADY wait-state	NONSEQ / SINGLE / WORD, HREADY pulsed low for 10 ns	0x6000_0000 / read

6. Running the Simulation

run.do (QuestaSim):

```
vlib work
vlog ahb_pkg.sv master.sv
test_ahb.sv
vsim -voptargs=+acc work.test_ahb
do wave.do
run -all
```

7. waveform snippet